

Henry Geerlings | Resume

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PhD candidate seeking full-time physical metallurgical/materials engineering position. Experience in materials characterization, foundry metallurgy, and data processing tools.

Education

Colorado School of Mines <i>PhD Materials Science</i>	Golden, CO 2021 - 2025
Colorado School of Mines <i>M.S. Materials Science</i>	Golden, CO 2016 - 2018
University of California <i>B.S. Materials Science & Engineering</i>	Berkeley, CA 2011 - 2015

Employment

Colorado School of Mines <i>PhD Candidate in Center for Advanced Non-Ferrous Structural Alloys (CANFSA)</i> Investigating the effects of residual copper on weldability and mechanical properties of low carbon steel sheet and plate. <u>Detailed Achievements:</u> - o Performed thermodynamic simulations to guide alloy/anneal design and predict laser weld hot-cracking susceptibility. - o Quantified differences in annealing, welding, and tensile behavior of steel sheet and plate as a function of copper content. - o Measured hot-cracking and cross-weld tensile properties for spot and laser welds. Characterized weld microstructures and fracture surfaces using optical and scanning electron microscopy. - o Senior foundry TA for sand and investment casting events, as well as the MTGN300L, "Foundry Metallurgy Lab" course.	Golden, CO Aug. 2021 - June 2025
CoorsTek <i>Materials Data Engineer in CoorsTek Center for Advanced Materials (CCAM)</i> Lead efforts within the Computational Materials group to assess and adopt best practices within R&D ceramics data. <u>Detailed Achievements:</u> - o Standardized data structure among multiple R&D programs in order to effectively share and reach target properties more effectively. Primarily investigated thermomechanical behavior of alumina and silicon nitride material systems. - o Developed and deployed high-throughput data processing tools to extract various features from historical datasets. - o Used Python and SQL to create and maintain various data pipelines for data fetching, processing, and modeling.	Golden, CO Aug. 2018 - July 2021
Colorado School of Mines <i>Researcher in Alliance for Development of Additive Process Technologies (ADAPT)</i> Defect and powder characterization in laser-based powder bed fusion additive processes. <u>Detailed Achievements:</u> - o Performed 3D μX-ray CT analysis of additively manufactured Inconel 718 samples and alloy powder stocks. - o Developed high-throughput routines for extracting, analyzing, and correlating porosity CT data with additive processing parameters and mechanical properties. - o Defined shape descriptors to quantify and compare porosity and powder particle "shapes" using image processing tools.	Golden, CO Feb. 2016 - June 2018

Publications & Conferences

- H. Geerlings, J. Klemm-Toole, A. Clarke, K. Clarke, **Effects of residual copper on the laser weldability of steel sheet**, *AWS Professional Program, FabTech Orlando*, 2024.
- H. Geerlings, J. Klemm-Toole, A. Clarke, K. Clarke, **Weldability of Scrap Plate and Sheet with Residual Copper**, *MS&T Conference, Columbus*, 2023.
- H. Geerlings, **TRACR: a software pipeline for high-throughput materials image analysis—an additive manufacturing study**, *Colorado School of Mines Thesis Repository*, 2018.
- M. Poschmann, J. Lin, H. Geerlings, I. Winter, D.C. Chrzan, **Strain-induced variant selection in heterogeneous nucleation of α -Ti at screw dislocations in β -Ti**, *Phys. Rev. Materials* 2. 2018.
- B. Kappes, S. Moorthy, H. Geerlings, D. Drake, A. Stebner, **Machine learning to optimize additive manufacturing parameters for laser powder bed fusion of Inconel 718**, *9th International Symposium on Superalloy 718 and Derivatives*, 2017.
- M. De Jong, W. Chen, H. Geerlings, M. Asta, K. Persson, **A database to enable discovery and design of piezoelectric materials**, *Scientific Data* 2, 2015.

Computing

OS: Windows, macOS, Ubuntu (Linux)

Utility: Git, DevOps, MS Office

Technical: ThermoCalc, FEniCS, Deform-3D, LAMMPS

Languages: Python, Bash, Matlab, SQL, $\text{\LaTeX}$

Equipment

Metallography: LOM, SEM, EBSD, EDS

Mechanical: Tensile, High Cycle Fatigue, Charpy

Thermal: Dilatometry, Box Annealing, Gleeble

Joining: Arc Weld Robot, Laser/Spot Welding, σ -Jig

Coursework and Miscellaneous

Materials Science and Engineering:

Crystallography, Bonding, and Defects
Phase Transformations and Kinetics
Mechanical Behavior of Materials
Experimental Materials Science
Thermodynamics in Materials
Strengthening Mechanisms
Materials Characterization
Forging and Forming
Solidification
Corrosion

Mechanical Engineering and Other:

Simulation of Advanced Manufacturing Processes
Engineering Analysis using FEM
Continuum Mechanics and Dynamics
Fatigue and Fracture
Computational Linear Algebra
Heat Transfer